

COMPUTED TOMOGRAPHIC FEATURES OF LUNG LOBE TORSION

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The imaging features of lung lobe torsion in 10 dogs (nine complete, one partial torsion) acquired with a helical single-slice computed tomography (CT) unit are described. Attenuation values of normal, rotated, and adjacent collapsed lung lobes before and after intravenous contrast medium administration were compared. Affected lung lobes were: left cranial (5), right middle (3), right cranial (1), and left caudal (1). CT findings in nine dogs with complete lung lobe torsion included pleural effusion and an abruptly ending bronchus. In eight of these dogs, enlargement, consolidation, emphysema of the affected lung lobe, and mediastinal shift to the contralateral side were present. Rotated lung lobes did not enhance, whereas adjacent collapsed and aerated lung lobes did ($P < 0.05$). Apnea induced with hyperventilation or breath-hold is essential to reduce motion artefacts and obtain a diagnostic study. *Veterinary Radiology & Ultrasound*, Vol. 49, No. 6, 2008, pp 504–508.

Key words: computed tomography, dog, lung lobe torsion.

Introduction

LUNG LOBE TORSION is a rare condition in dogs characterized by rotation of a lobe around its pedicle, leading to airway obstruction and vascular hypertension. It can develop spontaneously or as a complication of trauma, pleural effusion, or lobectomy.^{1,2} Lung torsion is associated with significant morbidity and mortality if it is not promptly corrected.³ It is therefore important to diagnose this condition early and accurately, and to differentiate it from other conditions with similar signs that do not require surgery. Because clinical signs, hematologic data and cytology of pleural effusion are not specific for lung lobe torsion; imaging is critical in making the diagnosis.⁴ Radiographic and ultrasonographic features of lung lobe torsion have been described, however, achieving an accurate diagnosis can be difficult with these modalities.^{5–9}

In humans, computed tomography (CT) has become the most valued tool to evaluate the structure of the pulmonary parenchyma and the thoracic cavity^{10–14} and CT is being used with increasing frequency in the assessment of thoracic diseases in dogs.^{15–17} The purpose of this study is to describe the CT protocols and imaging findings in dogs with lung lobe torsion.

Materials and Methods

Medical records between 2000 and 2007 from veterinary hospitals at the University of Pennsylvania and the University of Wisconsin—Madison, and the Clinica Veterinaria dell'Orologio were searched for dogs with a definitive diagnosis of lung lobe torsion. Only dogs that had a CT examination performed within 24 h before surgery were included.

Ten dogs were identified. Median age was 6 years (range 9 months to 10 years). There were two Pugs, and one each of Jack Russell Terrier, Akita Inu, Afghan Hound, Borzoi, English Bulldog, Beagle, Bernese Mountain dog, and Mix-breed dog.

Dogs were scanned in dorsal or ventral recumbency using a single-slice third generation helical CT unit (ProSpeed and HiSpeed Lxi, General Electrics, Milwaukee, WI). Expiratory apnea was obtained by hyperventilating the patient before the scan, and the scan direction was caudal to cranial. The most common protocol, used in eight of 10 dogs, included a survey helical CT imaging series using a collimation of 2–7 mm and a pitch between 1 and 2 with a medium frequency image reconstruction algorithm for soft tissues (proprietary term “detail”). Reconstruction interval was the same as collimation. In these eight dogs, manual cephalic vein injection of iodinated contrast medium at a dosage of 700 mg iodine/kg body weight (Hypaque[®]-76, Nycomed Inc., Princeton, NJ) was performed and post-contrast image series was acquired 1–2 min after contrast medium injection. In seven dogs, a high-resolution transverse CT imaging series with a collimation of 1 and 1–5 mm table increments, edge enhancement and a high frequency image reconstruction algorithm (proprietary term “bone”) was also obtained. In large breed dogs, either large table increments (5 mm) were used to cover the whole thorax during the expiratory pause, or the scan was limited to the

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area over the heart base. For the other two dogs, one had only a helical imaging series after contrast medium administration and in the other only precontrast transverse images were acquired.

Imaging protocols were retrospectively analyzed for image detail, motion artifact and quality of reformatted images. All images were retrospectively reviewed by one board-certified radiologist (G.S.) who evaluated all images for changes within the rotated lung lobe, changes within remaining lung and other changes within the thoracic cavity. The bronchial anatomy specifically was assessed using reconstructed images in various planes. Additionally, paddle wheel reconstructions,¹⁸ pivoting the reformatting plane around the bronchial bifurcation were performed to assess the course of the bronchial tree (Fig. 1).

In seven dogs with complete lung lobe torsion, attenuation values in the rotated lung were measured in Hounsfield units (HU) before and after intravenous (IV) contrast medium administration using regions of interest (ROI) with a size of 150–400 mm². The position of the ROI was identical in pre- and postcontrast images. ROIs were positioned over three different lung lobes: a normal, well aerated lung lobe; the rotated lung lobe and if present, an adjacent collapsed lung lobe. In the rotated lung lobe, ROIs were measured in three different slices to ensure

inclusion of more consolidated as well as more emphysematous areas. All three ROIs measured in rotated lung lobes were compared separately (not as a mean value for all three) pre- and postcontrast administration.

Differences between pre- and post-IV contrast medium administration images were compared using a paired Student's *t*-test (MEDCALC version 9.3.9.0, Mariakerke, Belgium). The level of significance was set at $P < 0.05$.

Results

Nine dogs had a complete lung lobe torsion. The affected lung lobe was the left cranial in five dogs (Afghan, Borzoi, English Bulldog, Bernese Mountain dog, and Pug), the right middle in two (Jack Russell Terrier, Beagle) the right cranial in one (Pug) and the left caudal in one dog (Akita Inu). One dog had a partial (180°) torsion of the right middle lung lobe (Mixbreed dog), determined in surgery. CT findings in dogs with complete lung lobe torsion are summarized in Table 1. Pleural effusion was present in all dogs with complete lung lobe torsion. The effusion was generalized in six dogs and focal around the diseased lung lobe in three dogs. The other CT finding that was present in all dogs was an abrupt ending bronchus in the diseased lung lobe. In three dogs a bronchial torsion could be seen (Fig. 2). In some dogs the bronchus was gradually thinned before the lumen was completely obliterated. Another frequent finding, seen in eight out of the nine dogs, was increased attenuation of the rotated lung lobe with areas of emphysema (Fig. 3). The affected lung lobe was increased in size in eight out of nine dogs (Fig. 3). None of the changes seen in rotated lung lobes were seen in unaffected lobes. In four dogs a deviation of the bronchi in the displaced lung lobe could be seen, and in two dogs the bronchi were compressed (Fig. 4). An abnormal position of the affected lung lobe was seen in five dogs (Fig. 5), in one patient only an abnormal course of the vasculature within that lung lobe, seen after IV contrast medium administration was present. Pneumothorax was seen in three dogs (Fig. 5), and one dog had mediastinal lymphadenopathy.

The dog with a partial torsion of the right middle lung lobe only had atelectasis of the affected lung lobe but a normal position and gas content of the bronchus (Fig. 6). Pleural effusion and pneumothorax were also present in this dog.

Contrast enhancement in normal inflated lung lobes and collapsed but not rotated lung lobes corresponded to a mean increase in attenuation of 56.7 and 87.4 HU, respectively, whereas mean enhancement in rotated lung lobes corresponded to a mean attenuation increase of 5.1 HU. Enhancement in rotated lung lobes was not statistically significant after contrast medium injection ($P = 0.12$) whereas the enhancement in adjacent collapsed ($P < 0.01$) and normal, inflated lung lobes ($P < 0.01$) was statistically



FIG. 1. Paddle wheel reconstruction computed tomography (CT) images of a 2-year-old pug with torsion of the right cranial lung lobe. From a transverse plane (top left corner) the reformatted plane is rotated around the bronchial bifurcation (top left to bottom right). All bronchi can be evaluated: a, left cranial; b, left caudal; c, right caudal; d, accessory; e, collapsed right middle; f, the very short, blind ending bronchus to the right cranial lung lobe.

TABLE 1. CT Findings in Nine Dogs with Complete Lung Lobe Torsion

Rotated Lung Lobe		Other Lung Lobes		Thoracic Cavity	
Findings	<i>n</i>	Findings	<i>n</i>	Findings	<i>n</i>
Abrupt ending bronchus	9	Lobar compression	8	Mediastinal shift	8
Enlargement	8	Displacement	6	Pleural effusion generalized	6
Consolidation	8	Deviated bronchi	4	Pneumothorax	3
Emphysema	8	Compressed bronchi	2	Pleural effusion focal	3
Abnormal position	5			Mediastinal lymphadenopathy	1
Twisted bronchus	3				
Abnormal course of vessels	2				

significant. The lung lobe with a partial torsion had a difference in attenuation of 58 HU, similar to the enhancement measured in normal or collapsed lung lobes.

Discussion

We identified a range of interesting CT features of canine lung lobe torsion, including pleural effusion, non-contrast enhancing single lung lobe enlargement with malposition, emphysema, and an abruptly ending lobar bronchus (Table 1).

The pathophysiology of lung-lobe-torsion-related pleural effusion is twofold. Torsion causes pulmonary venous hypertension, and decreased lymphatic drainage, both leading to pleural effusion.^{3,19} Pleural effusion is often the reason why radiographic interpretation is nonspecific in lung lobe torsion patients, whereas in CT images the effu-

sion does not represent a limitation. Pulmonary venous hypertension also causes enlargement of the affected lung lobe, together with development of emphysema which was seen in eight dogs of this series. Bronchial tears with a one-way valve effect are a possible cause for emphysema formation and pneumothorax.^{5,9} There is less evidence supporting the hypothesis that gas-forming bacteria cause the lobar emphysema.⁵ Pneumothorax may also be caused iatrogenically, due to thoracocentesis. In retrospect, it was not possible to determine the cause of pneumothorax in the dogs in this study.

An abruptly ending of the gas-filled bronchial lumen was observed in all dogs, together with an abnormal position of the rotated lung lobe and its bronchovascular structures in five of nine dogs. This is regarded as a convincing argument supporting lung lobe torsion. Abrupt ending of a bronchial gas-filled lumen can also be caused by neoplastic



FIG. 2. Dorsally reformatted computed tomography (CT) image of the thorax of an 8-year-old Afghan Hound with left cranial lung lobe torsion. The bronchus to the left cranial lung lobe ends abruptly (arrow), whereas a more distal segment is again air filled and dilated, giving the bronchus a twisted appearance.

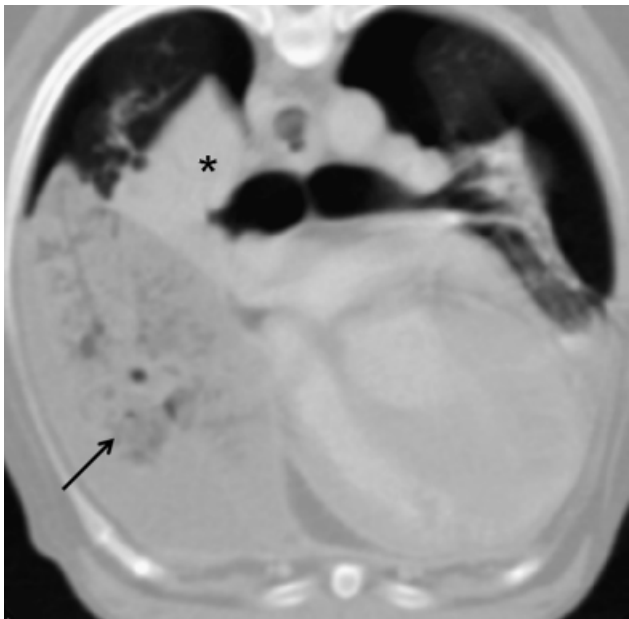


FIG. 3. Transverse computed tomography (CT) image of the thorax of an 8-year-old Jack Russel Terrier with torsion of the right middle lung lobe after intravenous contrast medium administration. The lung lobe is enlarged and contains trapped gas (arrow). The cardiac silhouette is displaced to the left. The right cranial lung lobe is dorsally displaced and compressed (*).

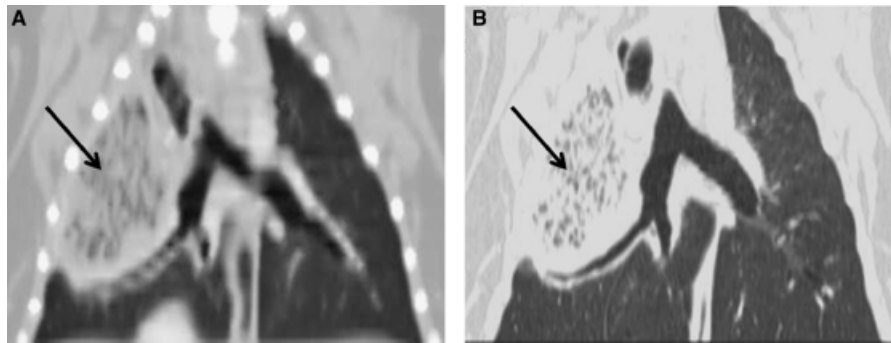


FIG. 4. (A) and (B): Dorsally reformatted computed tomography (CT) images of the thorax of a 2-year-old pug with torsion of the right cranial lung lobe. Helical 3 mm slices with a detail algorithm (A) and high resolution transverse 1 mm slices with a bone algorithm and edge enhancement (B) are shown. The lung lobe is enlarged and contains trapped gas. The bronchus to the right caudal lung lobe is displaced and curved around the enlarged cranial lung lobe (arrow). The right middle lung lobe is collapsed and medially displaced and cannot be seen on this image.

or other compressive or infiltrative peribronchial diseases.^{20,21} However in contrast to lung lobe torsion there is usually no vascular compression in these diseases leading to some enhancement after contrast medium administration. This is consistent with our findings, where completely rotated lung lobes did not have any significant contrast enhancement, whereas the adjacent atelectatic or aerated lung lobes did. Lack of contrast enhancement can also be seen in other necrotic, cystic, or thromboembolic conditions of the lung. However only parts of a lung lobe are usually affected in such instances. At an early stage of, or with less than complete lung lobe torsion, arterial blood supply can still be preserved, therefore presence of contrast enhancement does not completely rule out lung lobe torsion. Presence of arterial blood supply may be better seen

when imaging immediately after contrast medium injection during the arterial phase.

Several technical observations were made. Cardiac motion did not induce major motion artifacts. Hyperventilation-induced apnea was sufficient to achieve a motion-artifact free scan in most dogs. Using a caudal to cranial scan direction had the advantage of scanning the portion of the thorax with the largest amplitude of respiratory motion first if the patient started to breathe during the scan.

The pitch used was within the recommended range of 1–2.²² Ideally, the pitch should be about 1.4 to achieve the best compromise between image quality and scan speed.²² However a higher scan speed also increases the chance for a large area to be covered during apnea and thereby also contributes to image quality. In some patients, high-resolution CT was used (1 mm slice width, high-frequency

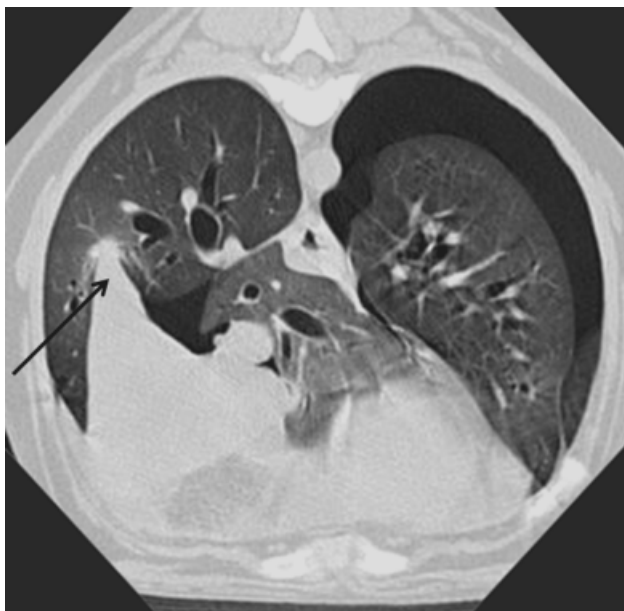


FIG. 5. Transverse high resolution computed tomography (CT) image of a Beagle with torsion of the right middle lung lobe. The tip of the lung lobe is pointing in a dorsolateral direction (arrow) indicating displacement and rotation of the lobe. Pneumothorax and mild pleural effusion are also visible.

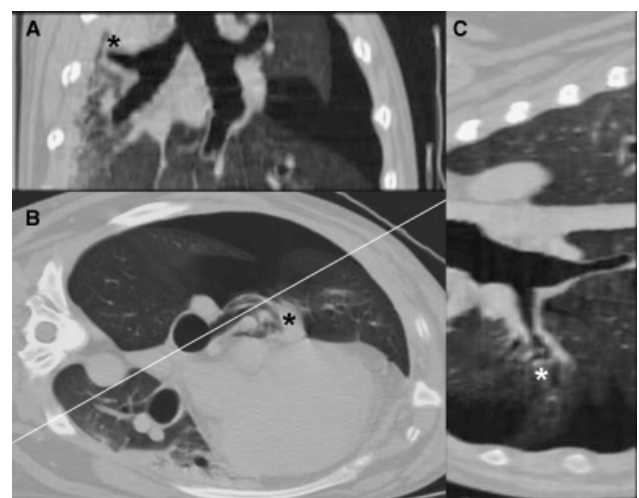


FIG. 6. Six-year-old mixbreed dog with partial torsion of the right middle lung lobe. Dorsally reformatted CT image (A), transverse image in left lateral recumbency (B) and obliquely reformatted image (C) in the plane indicated in Fig. 6B. The right middle lung lobe (*) is collapsed and the bronchus is curved, but in the reformatted images the entire bronchus is visible and air filled.

image reconstruction algorithm, additional edge enhancement, high mA, and lowest tube rotation time settings).²³ These images gave subjectively the best overall assessment of the pulmonary parenchyma and bronchi. The disadvantage of this thin slice technique is that only a small area can be covered during the 30–40 s of induced apnea. A combination of two thick-sliced helical series of the entire thorax pre-and postcontrast with one thin sliced high-resolution CT series restricted to the area over the heartbase may prove most successful to obtain motion artefact free images of the entire thorax and maximal anatomic detail for the lobar bronchi in the hilar area.

Especially helpful for assessment of position, course, and size of the bronchi was reformatting the images in various planes. The paddle wheel reconstruction method, pivoting the reconstruction plane around the bronchial bifurcation, was developed to facilitate evaluation of the natural course of the large bronchi and vessels.^{18,24} Because the center of rotation is in the center of the bronchial bifurcation, the large central structures appear on consecutive images and can be followed visually into the periphery of the lung (Fig. 1). Subjectively, reformatted images were most useful if the original series was acquired with thin slice thickness, as well as continuous axial slice interval or low helical pitch.^{25,26}

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